

Thyagarajulu Gollapalli

Research Officer and Software Engineer, Australian National University

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Research Interests

Computational geodynamics; geophysics; subduction dynamics; megathrust seismicity; mantle convection; fault representation in geodynamic models; numerical modelling; high-performance computing; reproducible open-source Earth-science workflows; geophysical data acquisition, processing, and interpretation.

Current and Past Work Experience

Research Officer/Software Engineer <i>Australian National University</i>	Jan. 2026–Present <i>Canberra, ACT</i>
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- Open-source developer for the Underworld geodynamics codebase, contributing core features, user-facing interfaces, testing, documentation, and reproducible modelling workflows.

Assistant Lecturer, Maternity Cover <i>Monash University</i>	Jun. 2025–Dec. 2025 <i>Melbourne, VIC</i>
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Senior Research Officer/Software Engineer <i>Monash University and AuScope</i>	Aug. 2023–Jun. 2025 <i>Melbourne, VIC</i>
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Imaging Geophysicist <i>CGG</i>	Nov. 2022–Jun. 2023 <i>Perth, WA</i>
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Junior Research Fellow in Gravity and Magnetism Lab <i>Department of Earth Sciences, IIT Bombay</i>	May 2017–Apr. 2018 <i>Mumbai, India</i>
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Project Assistant in Computational Geodynamics Lab <i>Centre for Earth Sciences, IISc</i>	Aug. 2015–Mar. 2017 <i>Bengaluru, India</i>
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Education

PhD, Computational Geodynamics and Geophysics Monash University and Indian Institute of Technology Bombay	Apr. 2018–Nov. 2022
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Thesis: *Unravelling tectonic coupling and loading along Sunda margin through 3-D regional numerical modelling*

Bachelor of Science, Mathematics; Minor in Earth and Environmental Science Indian Institute of Science, Bengaluru, India	Aug. 2011–May 2015
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Top Three Publications

1. Capitanio, F. A., **Gollapalli, T.**, Munukutla, R., Graciosa, J. C., Zuhair, M., Beall, A., & Dal Zilio, L. (2025). Bridging the gap between subduction dynamics and the long-term strength of megathrusts. *Nature Communications*. doi:10.1038/s41467-025-65824-7.
2. Moresi, L., Mansour, J., Giordani, J., Knepley, M., Knight, B., Graciosa, J. C., **Gollapalli, T.**, Lu, N., & Beucher, R. (2025). Underworld3: Mathematically self-describing modelling in Python for desktop, HPC and cloud. *Journal of Open Source Software*. doi:10.21105/joss.07831.
3. Graciosa, J. C., Capitanio, F. A., Beall, A., Hargreaves, M., **Gollapalli, T.**, Tang, T., & Zuhair, M. (2025). Testing driving mechanisms of megathrust seismicity with explainable artificial intelligence. *JGR: Solid Earth*. doi:10.1029/2024JB028774.

Awards

- Faculty of Science Dean's International Postgraduate Research Scholarship, Monash University, 2018–2022.
- Monash Graduate Scholarship, Monash University, 2018–2021.
- Postgraduate Publications Award, Monash University, 2022.
- AGU Fall Meeting Student Travel Grant, 2021.
- Junior Research Fellow Scholarship, Ministry of Human Resource Development, India, 2017–2018.
- INSPIRE Scholarship for Higher Education, Department of Science and Technology, India, 2011–2015.